

June 10, 2026

To the Editorial Board

Journal of Statistical Mechanics: Theory and Experiment (JSTAT)

Dear Editors,

Please consider the enclosed manuscript, “A two-feedback Vicsek–Couzin flock: density-stratified heading correlations as the robust fingerprint of motility–noise coupling,” for publication in JSTAT as an original research article.

The standard Vicsek model fixes the self-propulsion speed and the angular-noise distribution as global constants. Two relaxations have been studied separately: a density-dependent speed, which drives motility-induced phase separation, and heavy-tailed Lévy reorientations, which reshape the order–disorder transition. We ask what happens when both are switched on together, gated by a single density-dependent sigmoid in the local neighbour count, and we isolate each contribution with a four-way ablation (baseline / noise-shape-only / motility-only / full) run on one code path.

We find that heavy-tailed noise on its own grows no size-dependent susceptibility; only the motility channel builds a phase-separated dense phase, on top of which the otherwise-silent noise-shape adaptation adds a finite-size super-additive excess. The robust, scale-spanning fingerprint of the coupling is microscopic: a density-stratified dense-phase heading correlation $g(r)$ and its dynamic counterpart $C(\tau)$, together with a single-particle switch of dense-phase reorientation from Lévy-like to Gaussian (per-step turning-angle distribution and density-stratified mean squared displacement). A decoupled-sigmoid sweep identifies the motility threshold as the sole control variable, and a hydrodynamic appendix supplies the analytic counterpart.

We believe the work fits JSTAT squarely. It is a statistical-mechanics study of an active-matter model — Vicsek and Couzin flocking, motility-induced phase separation, giant number fluctuations, finite-size scaling of the order–disorder transition, and density-stratified correlation functions — and should interest the active-matter and non-equilibrium statistical-mechanics communities the journal serves.

The analysis is deliberately conservative. Claims rest on ten-seed finite-size scans with seed-bootstrap confidence intervals and non-parametric (Wilcoxon signed-rank, Mann–Whitney) tests rather than Gaussian z -scores; null results (a passive noise-shape threshold, negligible hysteresis, no extended patch lifetime, a topological-kernel breakdown) are reported rather than buried; and the macroscopic finite-size synergy is presented as a finite-size effect, not an asymptotic critical exponent. All code and data are openly available at <https://github.com/yayayou47/two-feedback-vicsek> for bit-for-bit reproduction of every figure.

This manuscript is original, has not been published previously, and is not under consideration at any other journal. It has a single author, who declares no competing interests. One supplementary-material PDF accompanies the submission. As potential referees, we would suggest specialists in active-matter and flocking models, in finite-size scaling of non-equilibrium phase transitions, and in motility-induced phase separation.

Thank you for considering our submission.

Sincerely,

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